ECE 329 **Homework 7 — Solutions**

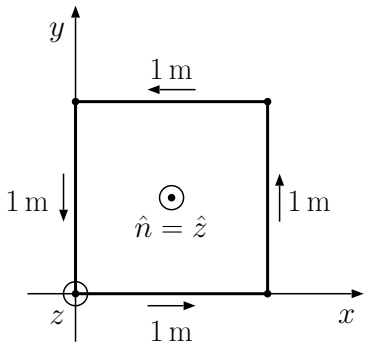
1. Since the closed paths are not varying in time and the magnetic field $\mathbf{B}$ is independent of position, we can rewrite Faraday's law as follows

$$\mathcal{E} = \oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d\mathbf{B}}{dt} \cdot \int_S d\mathbf{S} = \left(-\frac{d\mathbf{B}}{dt} \cdot \hat{n}\right) \cdot \text{Area},$$

where $\hat{n}$ is the unit vector normal to the surface, and

$$\frac{d\mathbf{B}}{dt} = B_0 \left((\sin(\omega t) + \omega t \, \cos(\omega t)) \, \hat{x} - \omega \sin(\omega t) \, \hat{z} \right).$$

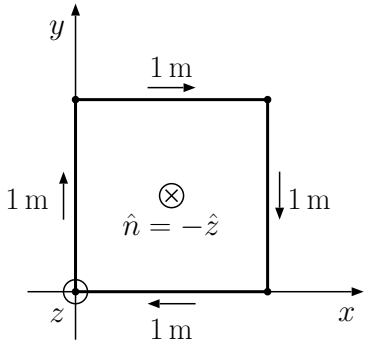
a) (3 pts.) For the rectangular path shown in the figure below,



the area of the enclosed surface S is $1\,\text{m}^2$ and the unit vector normal to S is $\hat{n} = \hat{z}$. Therefore, the electromotive force is

$$\mathcal{E} = -\frac{d\mathbf{B}}{dt} \cdot \hat{z} = B_0 \omega \sin(\omega t)\,\text{V}.$$

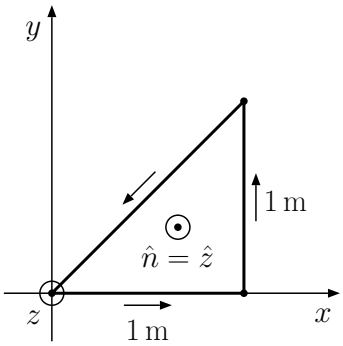
b) (3 pts.) If we consider the same rectangle of part (a), but the direction of the path is reversed, we have that $\hat{n} = -\hat{z}$.



As a result, the electromotive force is

$$\mathcal{E} = \frac{d\mathbf{B}}{dt} \cdot \hat{z} = -B_0 \omega \sin(\omega t)\,\text{V}.$$

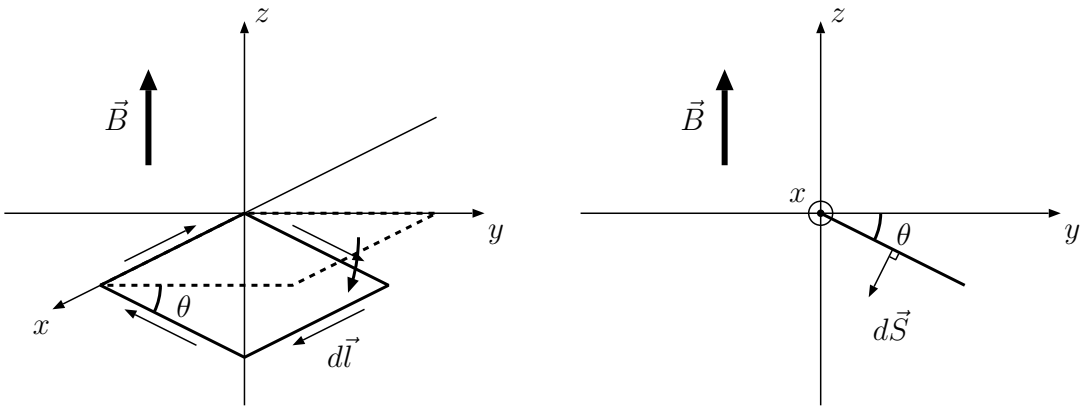
c) (4 pts.) For the triangular path shown in the figure below,



the area of the enclosed surface S is $\frac{1}{2}\,\mathrm{m}^2$ and the unit vector normal to S is $\hat{n} = \hat{z}$. Then, the electromotive force is

$$\mathcal{E} = -\frac{d\mathbf{B}}{dt} \cdot \hat{z} \cdot \frac{1}{2} = \frac{1}{2} B_0 \omega \sin(\omega t)\,\mathrm{V}.$$

2. Consider a square loop of wire of some finite resistance R with $1\,\mathrm{m}^2$ surface area that is located in a region of constant magnetic field $\mathbf{B} = 2\hat{z}\,\mathrm{Wb/m}^2$. The loop can rotate around the x-axis as shown in the following figure.



a) (2.5 pts.) If $\theta = 0°$, the differential area is $d\mathbf{S} = -\hat{z}dS$, and therefore, the magnetic flux is

$$\Psi = \int_S \mathbf{B} \cdot d\mathbf{S} = \mathbf{B} \cdot \int_S d\mathbf{S} = 2\hat{z}\,\frac{\mathrm{Wb}}{\mathrm{m}^2} \cdot (-1\,\mathrm{m}^2\hat{z}) = -2\,\mathrm{Wb}.$$

b) (2.5 pts.) For any angle θ, the differential area is $d\mathbf{S} = -\left(\cos\theta\hat{z} + \sin\theta\hat{y}\right) dS$, and therefore, the magnetic flux is

$$\Psi = \int_S \mathbf{B} \cdot d\mathbf{S} = 2\hat{z} \cdot \left(-\left(\cos\theta\hat{z} + \sin\theta\hat{y}\right)\right) = -2\cos\theta\,\mathrm{Wb}.$$

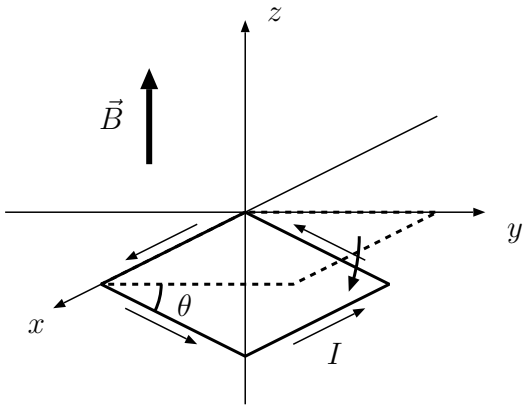
c) (2.5 pts.) The induced emf is given by $\mathcal{E} = \int_C \mathbf{E} \cdot d\mathbf{l}$. Hence, we write

$$\mathcal{E} = -\frac{d\Psi}{dt} = -2\sin\theta\frac{d\theta}{dt}.$$

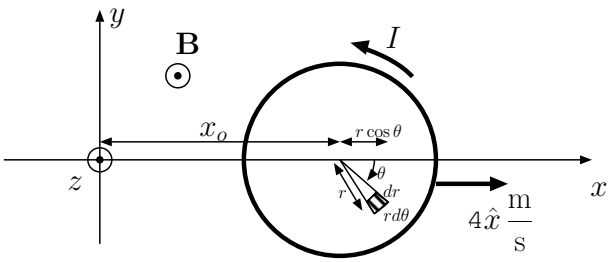
Considering $\theta = \frac{\pi}{4}\,\mathrm{rad}$ and $\frac{d\theta}{dt} = \pi\,\mathrm{rad/s}$, we get

$$\mathcal{E}(\theta = \frac{\pi}{4}) = -2\left(\sin\frac{\pi}{4}\right)(\pi) = -\sqrt{2}\pi\,\mathrm{V}.$$

d) (2.5 pts.) At the time when $\theta = \frac{\pi}{4}$, the induced emf $\mathcal{E}$ is negative, therefore, the current on the wire at the same instant flows in a direction opposite to the path of integration as it is shown in the figure below. We could have also found the direction of the induced current applying Lenz's rule (see the class notes).



3. The geometry of the problem is shown in the figure below.



a) (5 pts.) Looking at the described geometry, the magnetic flux is given by

$$\begin{aligned}\Psi = \int_S \mathbf{B}\cdot d\mathbf{S} &= \int_0^r \int_0^{2\pi} 25\times 10^{-6}(1+\frac{x_o + r\cos\theta}{L}) r d\theta dr, \\ &= 25\times 10^{-6}\left(1+\frac{x_o}{L}\right)\pi r^2\end{aligned}$$

Thus, the emf $\mathcal{E}$ is

$$\begin{aligned}\mathcal{E} = -\frac{d\Psi}{dt} &= -25\pi r^2\times 10^{-6}\times\frac{1}{L}\times\frac{dx_o}{dt} = -25\pi\times 10^{-6}\times\frac{1}{1000}\times 4 \\ &\approx -314.16\,\mathrm{nV}.\end{aligned}$$

b) (5 pts.) The magnitude of the loop current with resistance $4\,\Omega$ is

$$I = \frac{|\mathcal{E}|}{R} = 78.54\,\mathrm{nA}.$$

4. (10 pts.) According to Faraday's law, magnetic flux variations $\frac{d\psi}{dt}$ will cause a non-zero emf. Time derivative of the magnetic flux linking the square area S will yield

$$\mathcal{E} = -\frac{d\Psi}{dt} = 6\,\mathrm{V}.$$

Then, knowing that $V_1 = 2\,\mathrm{V}$, the voltmeter V2 will have

$$V_1 - V_2 = 6$$
$$V_2 = -4\,\mathrm{V}.$$

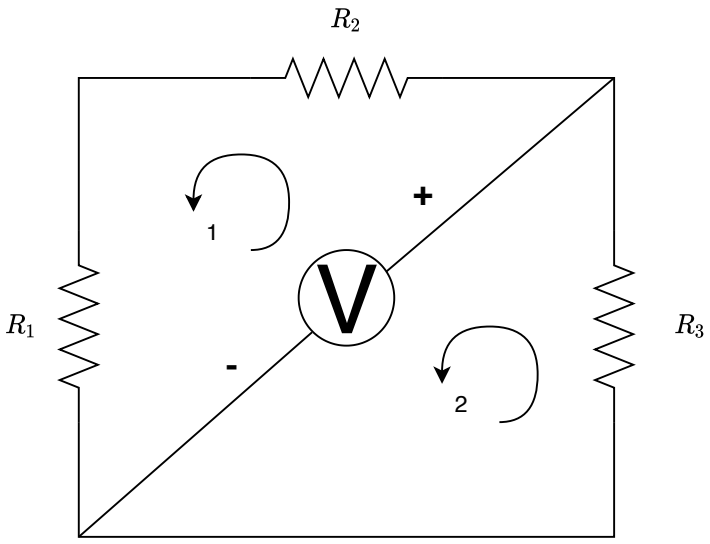
5. a) (2 pts.) According to Faraday's law, magnetic flux variations $\frac{d\psi}{dt}$ will cause a non-zero emf. Time derivative of the magnetic flux linking the square area S will yield

$$\mathcal{E} = -\frac{d\Psi}{dt} = 12\,\mathrm{V}.$$

b) (2 pts.) Using Ohm's Law, we have the loop current I as

$$I = \frac{\mathcal{E}}{R_1 + R_2 + R_3} = 2\,\mathrm{A}.$$

c) (4 pts.) The connection of the voltmeter is shown in the figure below.



The induced emf on loop 1 and loop 2 are both

$$\mathcal{E}_1 = \mathcal{E}_2 = -\frac{d\Psi}{dt} = 6\,\mathrm{V}.$$

Assuming the reading on the volmeter is V, we can find it using loop 1 as

$$\mathcal{E}_1 = I(R_1 + R_2) - V,$$

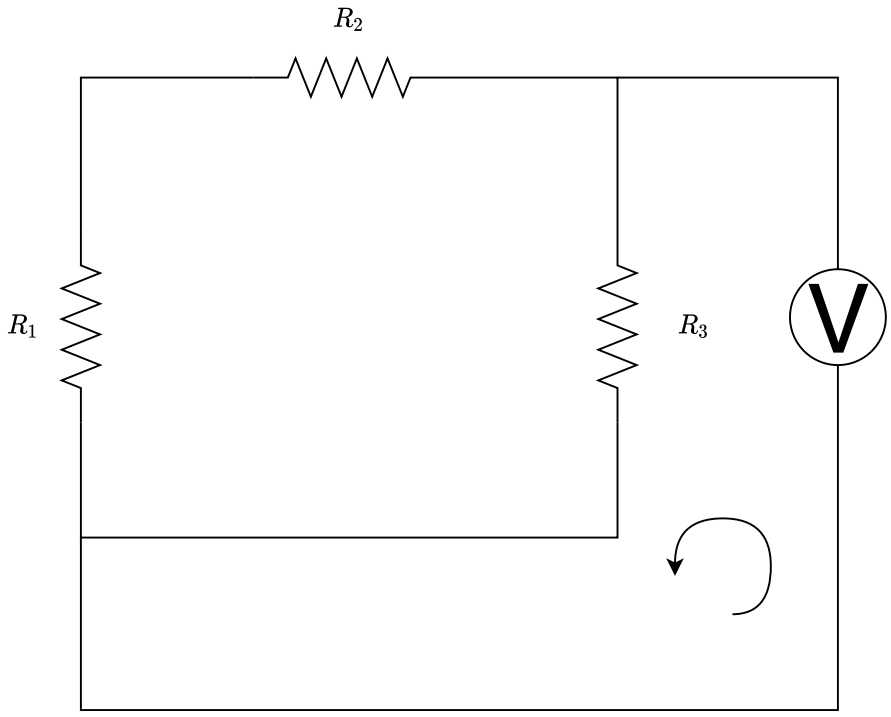
or using loop 2 as

$$\mathcal{E}_2 = IR_3 + V.$$

Both of them give us

$$V = 2\,\mathrm{V}.$$

d) (2 pts.) To avoid having the induced emf in the loop when we measure the voltage drop of R_3, we can move the voltmeter outside the area of the magnetic field as shown in the figure below.



The induced emf on the loop is now reduced to 0 and the voltmeter reading directly gives us the voltage drop on R_3.

6. (10 pts) Solution of the left-hand side of the Faraday's law $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ is given by

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & -\cos(\omega t + \beta x) & 0 \end{vmatrix} = \frac{\partial}{\partial x}\left(-\cos(\omega t + \beta x)\right)\hat{z} = \beta \sin(\omega t + \beta x)\hat{z},$$

which must be equal to $-\frac{\partial \mathbf{B}}{\partial t}$. Thus, we write

$$\beta \sin(\omega t + \beta x)\hat{z} = -\frac{\partial B}{\partial t}.$$

Integrating both sides of the above equation and dividing by $\mu = \mu_0$, we obtain

$$\begin{aligned} \mathbf{H} &= \frac{\beta}{\omega \mu_o}\cos(\omega t + \beta x)\hat{z} = \frac{1}{c\mu_o}\cos(\omega t + \beta x)\hat{z} \\ &\approx 2.65 \times 10^{-3}\cos(\omega t + \beta x)\hat{z}\,\frac{\mathrm{A}}{\mathrm{m}}. \end{aligned}$$

7. (10 pts) Ampere's law $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ is reduced to $\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$ since $\sigma = 0$ yields $\mathbf{J} = \sigma \mathbf{E} = 0$. Solution of the left-hand side of the Ampere's law is given by

$$\nabla \times \mathbf{H} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin(\omega t - \beta z) & 0 & 0 \end{vmatrix} = \frac{\partial}{\partial z}\left(\sin(\omega t - \beta z)\right)\hat{y} = -\beta \cos(\omega t - \beta z)\hat{y},$$

which must be equal to $-\frac{\partial \mathbf{D}}{\partial t}$. Thus, we write

$$-\beta \cos(\omega t - \beta z)\hat{y} = \frac{\partial D}{\partial t}.$$

Integrating both sides of the above equation and dividing by ϵ, we get

$$\begin{aligned} \mathbf{E} &= -\frac{\beta}{\omega \epsilon}\sin(\omega t - \beta z)\hat{y} = -\frac{3}{2c \times 2.25\epsilon_0}\sin(\omega t - \beta z)\hat{y} \\ &\approx -250.98\sin(\omega t - \beta z)\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}}. \end{aligned}$$